Structuring Interviews with Experts During Knowledge Elicitation

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We have developed interviewing techniques for knowledge elicitation based on (1) relevant research and techniques from the social sciences, (2) the nature of expertise, (3) a desire to assist a knowledge engineer to avoid reductive bias, one of the pitfalls associated with the acquisition of highly complex concepts, and (4) the desirability of de-coupling elicitation from implementation. The approach consists of four phases with guidelines for questions suited to each stage. First, the descriptive elicitation stage is intended to reveal the important entities and concepts in the domain as reflected in the terms and specialized language used by the expert. A second stage, structured expansion, is designed to probe the relationships between the domain concepts and the organization of the expert's knowledge using the terminology uncovered in the previous stage. The scripting phase relies on the declarative knowledge found through the two previous stages to discover procedural knowledge, and the final component, validation, is important throughout the process of knowledge elicitation to ensure that the knowledge being elicited is correct and adequate to enable a system in which it is implemented to solve the class of problems with which it is concerned. © 1993 John Wiley & Sons, Inc.

I. INTRODUCTION

Recently, Shaw and Woodward[1] proposed a framework in which knowledge acquisition methods and issues can be addressed in a systematic fashion. Their proposals are a reaction to their characterization of the field of knowledge acquisition as a methodological "patchwork," resulting from the importation and adaptation of methods from other disciplines. We have attempted to use Shaw and Woodward's framework as an organizing scheme in the present article to report our own work in developing structured interviews for the elicitation phase of knowledge acquisition. At the outset, we would like to clarify the use of the terms knowledge "acquisition" and knowledge "elicitation," following the distinction made by Motta *et al.*[2]. Knowledge "acquisition" will refer to the broader activity of gathering information from a variety of sources, one of which is a domain expert, and interpreting and organizing it. The activities

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involving direct interactions with an expert will be referred as knowledge "elicitation." We make this distinction because the work reported herein is concerned primarily with the latter activity.

The concerns expressed by Shaw and Woodward[1] about the field of knowledge acquisition, generally, have been expressed by Forsythe and Buchanan[3] about interviewing techniques for knowledge elicitation, more specifically. They point out that "while recent publications have come closer to addressing some of the practicalities of interviewing for knowledge acquisition, they still suffer from the general tendency to name methods without providing much information on how to apply them. . . . In short, we know a good deal about how to represent and manipulate knowledge in expert systems, but have very little systematic understanding of how best to gather that knowledge in the first place" (pp. 1–2). They then proceed to report some of their work designed to address this shortcoming. Whereas the issues addressed by Forsythe and Buchanan are directed at the social context and communication issues surrounding an interview with a domain expert, the work reported here is concerned with particular kinds of questions asked during an interview and with guidelines for when and how to ask them.

II. GUIDING PRINCIPLES AND ASSUMPTIONS

Although our work is centered around interviewing techniques, we do not mean to imply that we consider interviews to be adequate and sufficient for meeting all the needs of knowledge elicitation. Rather, our goal is to provide guidelines for making interviews with an expert as productive as possible. Also, as will become apparent, our conception of what constitutes interviewing is reasonably broad. Because our approach involves the importation and adaptation of methods from several disciplines, we attempt to provide theoretical bases for them grounded in those disciplines. Also, in keeping with the recommendations of Shaw and Woodward, we discuss the working assumptions and principles that have guided our work before describing specific aspects of our techniques.

A. The Nature of Expertise

It is not uncommon for a report on knowledge acquisition to begin with a statement to the effect that knowledge acquisition continues to be the major bottleneck in the process of building knowledge-based systems. As suggested by LaFrance,[4] one reason for this is that developers continue to underestimate the complexity of expertise. Indeed, this is consistent with our own experience and assessment of the knowledge acquisition literature. Thus one of the guiding principles underlying our approach to knowledge elicitation has been to consider the nature of expertise as it has been characterized by researchers in the discipline of cognitive psychology. As a result, we will briefly review some of the characteristics of expert knowledge that we feel should be considered by

those involved in knowledge acquisition. For a more thorough treatment of this issue consult LaFrance.[4]

Four aspects of expertise which we find particularly relevant to knowledge elicitation are the organization of knowledge, its influence on the representation of problems, characteristic problem-solving strategies, and the tacit nature of the knowledge. In addition to being sensitive to these four aspects of expertise, our methodology attempts to avoid "reductive bias," one of the pitfalls of advanced learning in complex domains.

1. Knowledge Organization

Experts have a great deal of domain-specific knowledge, compared to novices. However, more than quantity of knowledge, it is the qualitative differences that have implications for knowledge elicitation.[4] At a macro level, a variety of studies (e.g., Chi *et al.*[5]) have found expert knowledge to have a generally hierarchical organization (i.e., categories with subcategories), although many other types of relationships are also present, depending on the particular domain. All of this domain-specific knowledge is tightly interconnected with many associations among items. By contrast, novices do not yet have their domain-specific knowledge nearly as well organized or tightly integrated.

At the micro level, expert knowledge is stored as organized "chunks" of frequently occurring patterns or schemas.[4] In the case of chess, for example, Simon and Gilmartin[6] have estimated that grand masters have acquired on the order of 50,000 different patterns or chunks of chess information, each of which is associated with one or more strategies used in play. An important implication of this chunking or schema formation is that it enables an expert to quickly recognize patterns in the information encountered during problem solving. Part of schema formation is the association of functional characteristics and action sequences relevant to that particular pattern.

While expert knowledge has abstract dimensions, LaFrance[4] points out that it is important to recognize that experience is grounded in specific situations or "cases" that have been encountered in the past. While abstractions obviously arise from those specific cases, we must not lose sight of the fact that they remain an important part of the memory organization of an expert and provide an important basis for problem solving.[7]

2. Problem Representation

Differences in amount and organization of knowledge naturally lead to differences in the way novices and experts represent problem information as they analyze it during problem solving. For experts, representation generally takes the form of underlying principles and abstractions.[5,8] This is related to the notion of schemas discussed above. In contrast, novices tend to focus on superficial characteristics of a problem and objects described in the problem statement. In computer programming, for example, problem representation of

experts takes the form of language-independent, abstract constructs, whereas novices conceptualize problems in terms of commands and control structures specific to the programming language in which they are working.[9]

3. *Problem-solving Strategies*

Some of the characteristic ways in which experts approach problems carry across domains, while others are domain specific. One general approach is a working-forward strategy, which means moving from the information given in the problem to the goal via operations that the expert knows will work. In contrast, novices tend to begin at the goal, noting information that is missing. They then work backward one step at a time, determining at each point what is needed to solve the immediately succeeding step. Once all missing information is found, the steps can be reversed to solve the problem. This is particularly clear, for example, in the way people approach geometry proofs[10] and physics problems.[11]

Researchers have also documented a number of domain-specific problem-solving strategies. For example, in medical diagnosis experts develop the ability to entertain several competing hypotheses about the disease simultaneously, whereas novices tend to consider only one at a time.[12] This seems directly related to the larger degree of integration found in the experts' knowledge about diseases and symptoms and their relationships. Another example of domain-specific problem-solving strategies comes from computer programming. In this case experts take a top-down approach to program development, working at a global level and developing the program in a breadth-first manner. In contrast, novices tend to focus on low-level details in what is referred to as a depth-first strategy.[9]

4. *Tacit Knowledge*

It is a common observation that much of an expert's problem-solving knowledge has become automatized or tacit through extensive use, and there has been considerable debate about its accessibility, even to the expert.[13,14] Anderson's[15] account of the development of automatized knowledge describes this difficulty. According to Anderson, declarative knowledge includes the conceptual entities (both concrete and abstract) in a particular domain and their various relationships. Procedural knowledge, on the other hand, is comprised of the knowledge required for solving problems using relevant domain concepts and relationships. Another way to describe this distinction is a knowledge of "what" vs a knowledge of "how."

In early stages of skill learning an individual consciously considers various items of declarative knowledge during problem solving. In well learned tasks, much declarative knowledge is no longer consciously available during problem solving, although this does not mean that the declarative knowledge has been "forgotten." It does mean that it may be difficult for an expert to articulate it to a knowledge engineer, especially when asked to do so directly.

B. Reductive Bias

Because a knowledge engineer is often a relative novice in the domain of knowledge being elicited, there exists a potential for errors stemming from what has been termed "reductive bias." This is a problem that arises in situations involving the learning of complex bodies of knowledge, such as medicine. For example, Feltovich *et al.*[16] have researched the basis for significant, pervasive misconceptions that occur in medical students as they study to become specialized practitioners. They describe a particular case in which a serious misconception about the basis of congestive heart failure was found to exist not only in the minds of medical students, but also in the minds of some medical practitioners, and in some medical textbooks.

Feltovich *et al.*[16] outline characteristics of advanced "knowledge acquisition" that make unusual demands on cognition which tend to promote reductive (or simplification) biases. These, in turn, lead to misconceptions on the part of the student. Two examples of reductive bias are analogy bias and common connotation bias. The former results from the inappropriate application of an analogy that is too simple, or the inappropriate extension of attributes (e.g., if A and B are alike on attribute X, they are also alike on attributes Y and Z) Common connotation bias occurs when everyday terms are used in technical ways, but not understood (by the student) for their technical meaning.

In attempt to minimize the potential effects of reductive bias, we have imported techniques from the disciplines of anthropology and ethnography because researchers in those disciplines face a similar problem when they are attempting to elicit cultural information from informants (or cultural "experts") in unfamiliar societies. Spradley[17] makes a convincing case for the importance of language in this exercise. He notes that this can be particularly critical when the members of the culture appear to speak the same language as the researcher because important semantic differences exist that are not readily apparent. This is essentially the "common connotation bias" of Feltovich *et al.*[16] referred to above.

Spradley[17] describes another potential source of bias, which he terms "translation competence." This arises when informants translate the reality of their culture in order to explain it to an outsider. The problem is that the more an informant translates for the convenience of an investigator, the more that informant's cultural reality becomes oversimplified and distorted.

In an effort to avoid translation bias, Spradley[17] developed an approach to questioning that makes minimal assumptions about informants' knowledge and uses information they provide as the basis for further questioning. The researcher first uses very general probing techniques to persuade the informant to talk freely about his or her domain in a global sense. The informant's language is recorded and examined for category labels and other linguistic cues. This information is then used to cue the informant for additional information. Throughout the process it is necessary to verify that the researcher's emerging picture of the domain accurately reflects the informant's cultural expertise. We have adopted this general framework in which to develop a knowledge elicitation methodology.

C. "Unlocking" Tacit Knowledge

While direct access to automatized knowledge may not be possible, an assumption behind our approach is that it is possible to use information about how the expert views the domain to constrain inferences about tacit knowledge. In essence, it involves the systematic use of declarative knowledge to elicit procedural knowledge.

To illustrate this point, we turn to the domain of telephone interviewing, in which one of the authors has done knowledge modeling.[18] The observable data from the interviewing task consist of a verbal transcript of the interaction between a skilled interviewer and various respondents. Careful investigation reveals that interviewing expertise includes a classification scheme of questions, respondents, and response deficiencies. An incautious approach to knowledge elicitation that attempted to relate the nature of specific respondent statements to specific interviewer probes without reference to the abstract categories used by interviewing experts would be inadequate in this domain. If, on the other hand, the knowledge engineer understands the expert's way of thinking and talking about the domain, patterns can be sought at a more useful level of abstraction. An awareness of the declarative knowledge used by experienced interviewers enables the knowledge engineer to make sense of the observable data of the telephone interviewing task. Thus accurately capturing this procedural knowledge is dependent on a clear conception of the distinctions inherent in the categorizations used by the expert.

D. De-coupling Elicitation and Implementation

Our approach is heavily influenced by our general belief that knowledge should be elicited, organized, and documented with minimal concern about constraints and details associated with implementing a working system in which it will be used. The major advantages of such an approach are that it does not make commitments to tool-dependent representations (e.g., rules, frames, predicate logic, etc.) and it can yield an understandable means of communication among knowledge engineers and experts. This position, which has been advocated by others,[19–21] is particularly important when the knowledge domain is abstract or ill-structured, thereby requiring more extensive domain definition and conceptualization. It is also useful when the knowledge is to be used in an instructional system, for example, because of the need to analyze and restructure knowledge to optimize learning.

III. INTERVIEWING STRATEGIES

We have developed a four-phase approach to knowledge elicitation which addresses the issues outlined above. As indicated earlier, this approach draws heavily upon ethnographic methods, but also makes use of techniques from cognitive psychology and the current knowledge elicitation literature. The four phases are (1) Descriptive Elicitation, which focuses on helping the expert to

provide a large quantity of his or her declarative knowledge about the problem-solving domain; (2) Structured Expansion, which has as its focus the expansion and integration of the fragmented portions of the expert's declarative knowledge obtained in Descriptive Elicitation; (3) Scripting, in which the focus changes from obtaining declarative knowledge to using this declarative knowledge to constrain inferences about the expert's "compiled" procedural knowledge; and (4) Validation, which is concerned with the validity and psychological reality of the knowledge engineer's evolving conception of domain problem solving.

We certainly do not intend the order of presentation to imply that knowledge elicitation can or should be performed in a rigid order. Techniques from all four phases (particularly Validation) may be useful at any time in the elicitation process. However, as suggested earlier, we do rely on making the conceptual structure of declarative knowledge explicit, and using it to guide the elicitation of procedural knowledge (see also de Greef and Breuker[22]).

A. Descriptive Elicitation

In Descriptive Elicitation, the knowledge engineer attends to the categories, objects, models, and other conceptualizations used by experts in describing their domain. Because much of this information is reflected in the expert's use of domain-specific terminology, a significant amount of the knowledge engineer's time in this phase should be spent documenting the expert's use of language. The most important considerations during this early phase of knowledge elicitation are to record verbatim what the expert says and to ensure that the terminology being used is "native" to the expert's community of specialists. It is important to remember that a word or phrase need not be obviously unfamiliar to the knowledge engineer in order to be important. In the domain of telephone interviewing, for example, the term "TQ" has no meaning to a novice and would be an obvious point of focus. What might not be obvious, however, is that the terms "survey," "interviewer reject," and others which outsiders may find meaningful have special significance in this domain. The knowledge engineer should attend to all terms and phrases that are used frequently by the expert.

Like many social science investigative techniques, Descriptive Elicitation is based on the use of indirection. The expert concentrates on providing examples and descriptions of different aspects of the domain, while the knowledge engineer attends to the terms and labels that represent domain-specific abstractions. This is an attempt to counteract the possibility that an expert will reflect on automatic processes to such an extent that they will be interrupted or distorted. The goal of Descriptive Elicitation is to direct the expert's attention toward the complex task of describing the structure of the problem-solving domain. It is important for the knowledge engineer to attempt to create a situation in which the expert will describe the domain in a natural way, using domain-specific labels for important concepts. However, the knowledge engineer must also be concerned about keeping the expert focused on that portion of the domain which has been targeted for knowledge elicitation. Otherwise,

Table I. Descriptive Questions.

Question Type	Example
Grand Tour Questions	
Typical Grand Tour	"Could you tell me about a typical telephone interview?"
Specific Grand Tour	"Tell me about the last interview you did, from the time you picked up the survey form until you turned it in to your supervisor."
Guided Grand Tour	"Could you turn through the survey form and point out the different parts?"
Task-Related Grand Tour	"When you fill out the Call Record Sheet for the next interview, could you tell me about each step?"
Minitour	"Tell me about a typical open-ended question?"
Case-Focused Questions	
Example	"Can you give me an example of 'probing for a more specific response'?"
Personal Experience	"You've probably had some interesting experiences dealing with 'flakey responses.' Tell me about some of them."
Second-Hand Experience	"What kinds of experiences have you heard other telephone interviewers tell each other about 'ambiguous responses'?"
Native-Language Questions	
Direct Language	"What would you call a respondent who kept saying he was going to hang up?"
Hypothetical-Interaction	"How would you describe what you just told the respondent if you were telling your supervisor about it?"
Typical Utterance	"What are some sentences or phrases I might hear interviewers use that contain the term: 'TQed'?"

the interview can become very inefficient. Specific types of questions are designed to assist the knowledge engineer to meet these various goals.

One way to elicit a large sample of expert terminology is to ask a number of "Grand Tour" questions, which encourage the expert to verbally "show the knowledge engineer around" the physical, temporal, and abstract space of the problem-solving domain. Examples of such questions are shown in Table I. Grand Tour questioning is of particular utility early in the elicitation process, but may be used whenever the expert identifies a new subproblem. Asking about typical problem-solving episodes often elicits general categories used in frequently occurring situations. Some experts, however, may be uncomfortable with this degree of abstraction. Such situations call for the use of Specific Grand Tour questions, which refer to more concrete situations.

Case-Focused questions are also useful in eliciting domain terminology. As indicated earlier, cases are indexed in memory by the categories of which

they are members. These types of questions elicit further terminology used to characterize domain problem-solving by evoking the expert's description of the characteristics and details of particular problem-solving episodes. They also provide a basis for keeping an interview focused on relevant topics, especially if an expert tends to get sidetracked.

Example questions help the knowledge engineer to obtain more detail about terms already identified. Personal Experience questions are also useful for this purpose, but can be a source of distraction if used too early in the elicitation process. The most useful domain abstractions for early knowledge engineering efforts are those that are relevant to many potential cases in a domain. Personal Experience questioning may elicit "war stories," which are memorable to the expert because of their atypicality. Because terms and abstractions unique to such atypical cases are useful in modeling uncommon problems, Personal Experience questions are more useful later in the elicitation process, when knowledge relevant to special cases is being added to the evolving knowledge base.

Second-Hand Experience questions also suffer from the atypicality problem, but provide the knowledge engineer with indirect access to the larger "community" of domain experts. Communication between experts within this community about domain problems and knowledge is one of the processes that fosters the development and use of specialized domain terminology.[17] Accounts of cases encountered by other experts will likely have been discussed because of their interest to the community of experts in general. If two or more experts are readily available for knowledge engineering, it is useful to observe them as they discuss various cases.[23]

Although Descriptive Questioning is designed to use indirection, there will undoubtedly be times when this strategy will fail. As discussed earlier, Spradley[17] warns that cultural experts tend to translate their knowledge into terms they believe the interviewer will find easier to understand. Native Language questions can be used when this is suspected. Although it is difficult to tell exactly when an expert is translating, there are some cues. The use of analogies to everyday objects and situations, formal explanations from texts, or other explanations that the expert takes time to construct are not likely to represent the expert's customary modes of thought. Although such explanations may be useful at some point in the elicitation process, they interfere with early attempts to obtain samples of language relevant to natural domain reasoning.

One technique often used by ethnographers[17] to elicit informants' natural descriptions of terms and concepts is to ask how terms, tools, and concepts are "used" rather than what they "are" or "mean." Questions about meaning tend to encourage the expert to translate. Queries about use prompt the informant to describe the context in which a particular term or object plays a role. This description provides information that can be used to structure more in-depth questioning. Typical use-oriented questions are "When might I hear someone say 'interviewer reject'?" or "What is a 'call record sheet' used for?"

B. Structured Expansion

In Structured Expansion, the knowledge engineer uses questioning techniques that explore the rich, integrated organizational structure of the expert's knowledge. These techniques are based on the general principle that concepts in memory are associated with one another, that remembering or being cued with one or more domain concepts will result in the recall of additional relevant concepts. This basic principle underlies several general guidelines for questioning experts, and enables the knowledge engineer to construct rich, schematic structure of expertise described earlier.

The first technique, and perhaps the most important, given our focus on language, is to phrase questions to the expert in domain-specific terminology. A knowledge engineer's consistent use of terminology different from that of the expert will encourage the expert to translate. Using the domain-specific language, however, provides a context of familiarity and encourages the expert to focus on the domain itself rather than the knowledge engineer's methods and unfamiliarity with the domain. The expert is more likely to retrieve domain-relevant information in response to domain-relevant cues than nonrelevant ones.

There are several additional ways to cue experts to provide information while minimizing bias. One technique used by linguists[24] is to simply increase the length of the questions. Experience has shown that when even the simplest question is accompanied by an explanation, an expression of appreciation, or any nonleading discussion, then the interviewee will provide a longer response. In addition to being longer, the response is less likely to be distorted. When an expert answers a brief direct question, responses are made to those associations in memory evoked by the cues in the question. On the other hand, when an expert provides a prolonged response or description, there is less direct influence from the interviewer.

When possible, it is preferable to elicit knowledge in the setting in which the expert usually solves problems.[25] The familiar surroundings serve as further cues to the knowledge used by the expert in domain problem solving. The questioning context also has an effect on the amount of information provided by the expert. Similar questions should be asked in several contexts to elicit complete information. Also there is evidence[26,27] indicating that individuals select a limited number of all possible associations for report in any given interviewing context. It would be premature to conclude that knowledge about a given term or topic is "tacit" and completely unavailable to an expert when he or she fails to respond in a single questioning context. Rather, it is advisable to extend questioning about each term over a number of interviewing contexts in order to maximize the total amount of information retrieved.

These considerations are important, but not sufficient alone to guide further elicitation efforts. Unstructured free association by itself is not a productive method for building models of expert knowledge. Additional information about the form or content of the expert's knowledge is necessary if questioning is to produce a coherent and organized model of that knowledge. Ethnographers have developed methods for eliciting hierarchies of concepts based on language

Table II. Structural Questions.

Question Type	Example
Relationship Questions	
Cover Term	"Are there different kinds of 'respondents'? Tell me some of the different kinds."
Included Term	"Are 'repeating the question,' saying "Tell me more about that," and saying "That's interesting" all one kind of thing?"
Substitution Frame	"How many different ways can you think of to complete the sentence 'Respondents can cause problems for interviewers by _____.'?"
Contrast Questions	
Directed Contrast	"Could you look through this interview and show me what other questions are often seen as 'sensitive'?"
Dyadic Contrast	"Do you see any differences between 'statisticals' and 'open-ended' questions?"
Triadic Contrast	"Which two of the three responses written on this page are alike and which one is different from the other two?"
Contrast Set Sorting	"Please sort these cards [containing written probes] into two or more piles, depending on how they are alike or different."
Rating/Ranking	"What types of questions do you like best and which are your least favorite?"

cues. Examples of questions based on these principles are presented in Table II. The knowledge engineer needs to look for "cover terms" or terms denoting what we would call concepts. Each concept may be a cover term associated with a number of "included terms," each of which are related to the cover term by a particular semantic relationship. All of the included terms related to a given domain concept via the same relationship constitute a "contrast set" among which meaningful distinctions are made.

Suppose, for example, the expert said, "Well, there are different kinds of surveys, such as political surveys and marketing surveys." This indicates that there are at least two included terms in the domain defined by the cover term "surveys" and the semantic relationship *kinds of*. These two entities are significantly alike in being *kinds of* surveys and it is likely that the expert can provide further information about other kinds of surveys and make meaningful distinctions among them. The most fruitful way to go about the task of eliciting such domain structure is to first identify the cover term and semantic relationship from a sample of expert language elicited during descriptive elicitation activities. Then the knowledge engineer should elicit as many included terms as possible by asking Cover Term questions (see Table II) or by having informants fill in Substitution Frame questions (see Table II) in as many ways as possible. Once the informant indicates that there are no more included terms in a particular domain, the knowledge engineer should elicit the relevant features and dimensions of contrast which discriminate the included terms from one another. This

is accomplished by asking a number of Contrast questions (see Table II). As before, it is important to attend to the expert's terminology for these dimensions and features.

The best place to begin building domain hierarchies may be with "basic-level" domain terms,[28–30] which can be tentatively identified by their frequency of use by the expert, their lexical brevity (note the prevalence of acronyms and other abbreviated forms in the jargon of many domains), and, to a lesser extent, by their unfamiliarity to the knowledge engineer. The use of Contrast questions and extensive general probes (see Table II) for more information at this level exploits the larger number of distinctions and associations to these key concepts in the expert's memory.

Boose and Bradshaw[31] state that the most profitable unit of analysis for knowledge base development is the "distinction," but they admit that attempting to elicit distinctions without some form of constraining context can be problematic. Confusion may result when experts are requested to create distinctions between elements at what the authors describe as different "levels of abstraction." Thus, if a knowledge engineer fails to give proper concern to domain structure, artificial distinctions may be elicited from the expert which are not relevant to problem solving.

Limiting requests for distinctions to contrast sets of included terms helps ethnographers avoid the problem of eliciting artificial distinctions. They have found that informants will give "test question responses" in an attempt to provide the "right" answer and will make artificial distinctions, resulting in an inaccurate portrayal of their knowledge. Conklin[32] illustrates this problem by describing various ways of segmenting the domain of "U.S. currency." In addition to a classification hierarchy that includes distinctions that users of U.S. currency make in practice (e.g., "coins vs cash"), Conklin also presents examples of artificial classification hierarchies that distinguish various units of currency according to physical features not normally used for discrimination (e.g., whether the president is facing right or left, what Latin words appear on the back, etc.). He notes that artificial classification hierarchies may initially appeal to the researcher because they adequately distinguish between their members, but that they do not contain the distinctions used in domain problem solving. Our focus on the structure of expert domain knowledge is an attempt to avoid this difficulty. Although we have emphasized categorical relationships in our discussion here, there will obviously be other important relationships that must be noted (e.g., part–whole, attributes, cause–effect), depending on the domain under study.

C. Scripting

As previously discussed, once the content and structure of a large portion of the expert's declarative knowledge have been elicited, this information can be used to guide the search for procedural expertise. We make a distinction between two different types of procedural knowledge: (1) the reasoning or mental manipulation of information an expert goes through to reach conclusions

about the problem at hand, and (2) the strategies and sources used by the expert to obtain the relevant information.

The most successful technique developed in cognitive science to investigate problem-solving strategies is think-aloud protocol analysis. The term "protocol analysis" represents a family of techniques used to examine subjects' verbalizations about their problem-solving processes. Ericsson and Simon[13] have observed that the analysis of a verbal protocol must be preceded by a domain definition phase, during which the researchers must exhaustively catalogue the labels for objects, concepts, etc., referred to in the protocol. The purpose of this phase is to provide researchers with the declarative knowledge necessary to interpret subjects' statements about the processes involved in problem solving.

The utility of using declarative knowledge to interpret verbal protocols is the reason we recommend obtaining extensive declarative knowledge before attempting to elicit procedural knowledge. Once the important concepts and objects in the domain are understood by the knowledge engineer, a think-aloud protocol is much easier to interpret. Because an extensive discussion of analysis is beyond the scope of the present article, we will focus here on think-aloud protocol generation. When emphasizing elicitation methods as we are doing here, methods for protocol generation simply become additional interviewing techniques, although they are usually not categorized with interviewing techniques in the knowledge acquisition literature (e.g., Shaw and Woodward[1]).

Reasoning strategies emerge from a variety of think-aloud protocols generated by asking experts to describe their thought processes as they solve specific problems or cases. As described earlier, one of the important aspects of expertise to be captured is the set problem-solving strategies used by the expert. The approach we are advocating here suggests more of a bottom-up as contrasted to a top-down approach (e.g., Breuker and Wielinga[20]) to capturing strategic knowledge. We do not wish to dichotomize this issue because it is obviously a continuum, and the two approaches should be seen as complementary. An approach more characteristic of the top-down variety would be the use of generic tasks (e.g., Chandrasekaran[33]), which are used to constrain knowledge acquisition efforts very early on.

A full discussion of the top-down/bottom-up issue is beyond the scope of this report. Here we simply want to point out that our approach is more toward the bottom-up end of the continuum, particularly at the outset. Obviously as more and more knowledge is obtained, it is used to guide and constrain subsequent elicitation efforts. Certainly, there are circumstances where both approaches are useful, and one of the issues in which we are interested is specifying those more carefully. On the surface, it would appear that situation characteristics that would argue for a top-down approach are a well defined target system, a reasonably concrete domain, or a domain that is well documented. On the other hand, a bottom-up approach would seem more appropriate where the target system is less clearly specified in the beginning (e.g., an intelligent tutoring system), the domain highly abstract, or there is little written documentation available. For research purposes, especially, we are concerned that a knowledge engineer might fail to find important information by overemphasizing a top-down approach.

Table III. Scripting Questions.

Question Type	Example
Protocol Generation Questions	
Think-Aloud	"Fill out the Call Record Sheet and say whatever comes to mind while you work. Do not worry about talking in complete sentences or 'making sense'."
Aided-Recall	"As we listen to this tape of the interview you just did, say whatever you remember about what you were thinking. Stop the tape or rewind whenever you want."
Cross-Examination	"What led you to terminate the interview with that particular respondent?"
Consultation Questions	
Expert/Client Interaction	"How would you go about deciding if another interviewer was dealing with a 'paranoid' or a 'slow' respondent?"
"Twenty Questions Game"	"Okay, I'm thinking of a kind of respondent. Ask me some questions and try and guess the type of respondent I'm thinking of."

There are at least three types of protocol-generating procedures that are useful in knowledge elicitation. As shown in Table III, Think-Aloud questions simply request an expert to report the contents of conscious awareness during the solution of a domain problem. Ericsson and Simon[13] caution that this procedure can only yield meaningful results when the task being performed concurrently with the protocol does not require the use of verbalization or other processes that tax the resources of working memory. Even when this condition holds, a subject is only likely to provide an incomplete account of the moment-to-moment contents of consciousness. Information that is heeded and processed automatically will not be reported at all. However, these results may still be valuable because they provide information about the sequence in which the expert attends to various problem features and the various subgoals for which these features are relevant.

Aided-Recall questions (see Table III) can be used to follow up results obtained in think-aloud protocols. This technique, which has been used successfully in eliciting descriptions of expertise,[34,35] involves video- or audio-taping the performance of an expert engaged in domain problem solving and then reviewing this record with the expert at a later time. This procedure frees the expert from the interference resulting from think-aloud protocols and creates a situation in which recall of processing is cued by an unbiased record of the expert's own performance. A variation on aided-recall, called cross-examination (see Table III), has been used successfully[36] to probe the limits of knowledge directly. After completing a think-aloud protocol on a particular problem, the expert is asked specific questions about it.

These methods can be supplemented by the knowledge engineer's use of

the previously elicited declarative knowledge to interpret and to fill in gaps in the protocols. When an expert mentions a particular concept or term, the knowledge engineer can use knowledge of the expert's conceptual framework to infer other relevant concepts which may be playing a role in decision making at that moment. The knowledge engineer is still required to do some speculating about the expert's reasoning processes, but these speculations will now be constrained by several different types of information.

A description of expert problem-solving knowledge is incomplete if it does not include information about how important problem information is obtained. "Expert/Client Interaction questions" (see Table III) can be used to discover which of several very general types of strategies the expert uses to obtain a particular piece of information. Experts obtain information from a number of sources: by direct perception of the problem situation, by asking questions of clients, from memory, reference materials, etc. If information is obtained from reference materials, by calculation or measurement, or by some other easily accessible means, no further elicitation from the expert on that topic may be necessary. Information that is directly perceived from the problem situation or indirectly elicited from clients may require additional attention.

Information that experts elicit from clients (when that is relevant) can be obtained by observing the consultation process. Naturalistic researchers seek out similar "query-rich" settings[37] in which their informants are naturally engaged in information elicitation of their own. Aided-Recall questioning can be used to prompt experts for the reasons underlying their various lines of questioning as outlined above. If the knowledge elicitation process has reached a point where the knowledge engineer can pose realistic problems to the expert, then it may be useful to engage the expert in a "Twenty Questions Game" (see Table III). The knowledge engineer organizes information about a specific problem and the expert must elicit this information from the knowledge engineer in order to determine a solution. This is one example of a "constrained information processing task" in which the expert's access to various types of information is restricted in order to determine more precisely what information is used to make various decisions. Hoffman[38] provides examples of how this strategy can be used to specify information which an expert obtains through direct perception.

D. Validation

It goes without saying that a constant concern during knowledge elicitation activities is to capture an accurate view of the domain and the procedures used by the expert to solve the class of problems under consideration. We divide validation activities into "checks," or attempts to validate knowledge in the possession of the knowledge engineer, and "controls," or methods designed to reduce error and misunderstanding in the knowledge elicitation process. Traditional case-based verification (i.e., trying new cases) is a kind of check, which may only occur very late in the knowledge engineering process and often concentrates on coding and implementation.[39] All of the elicitation methods we

Table IV. Validation Questions.

Question Type	Example
Native Language	"Would you describe this situation by saying 'the guy TQed in the statisticals' when talking to another interviewer?"
Domain	"Are there different kinds of 'probes'?"
Included Term	"Is a 'statistical question' a kind of 'closed-end' question'?"
Semantic Relationship	"Is it true that a 'chatty person' is a *kind of* respondent?"
Contrast	"Would it be accurate to say that you have to write out the answer to an open-end question while close-end questions have options to be circled?"
Sequencing	"The steps in asking an open-end question are (1) Read the question verbatim; (2) Write down the respondent's answer verbatim; (3) Decide if the response is adequate; and (4) Probe if the response is not adequate or go to the next question if it is. Is that right?"
Case	"Let's go through an interview and I'll play the role of an interviewer using the knowledge base to select the appropriate interviewer action at each step. Tell me when I do something that you would not."

have described above, however, can be classified as controls. We believe that following a careful and controlled methodology that minimizes bias can be at least as important as rigorous postelicitation checking. It is important to include both checks and controls throughout the knowledge elicitation process. The following general principles characterize effective use of checks.

Checks should be conducted during all stages of knowledge elicitation. The list of questions in Table IV contains sample check questions that apply to knowledge from each of the other three phases of elicitation. These and other similar questions should be used to verify every type of knowledge gathered during elicitation. Although our preference for conducting elicitation independently of implementation precludes the possibility of using expert evaluations of rapid prototypes, we do advocate expert critique of the knowledge engineer's ability to use domain knowledge. Case Verification questions in which the knowledge engineer solves domain problems and is evaluated by the expert provide the same kind of feedback.

Another type of check is appropriate when the expert uses analogies to communicate about the domain. One of the specific types of bias to which the knowledge engineer may be prone is analogy bias, where a single analogy in a relevant source domain is used that accurately captures some of the features of the target domain, but misses or distorts others (see Feltovich *et al.*,[16] as described above). The lesson this provides for knowledge elicitation is when an analogy is used by the expert, the knowledge engineer should explore it in depth to find exactly what aspects of the analogy apply and to what extent. When limitations are reached, the knowledge engineer might ask if there are other analogies that might apply to this segment of knowledge.

Evidence from cognitive psychology[40] as well as from the naturalistic tradition[41] suggests that people are prone to a systematic "verification bias" when validating knowledge. For example, in validating information such as "women are easier to interview on the telephone than men" there is a tendency to look for confirming evidence (e.g. hard-to-interview men and easy-to-interview women) rather than for disconfirming evidence (e.g. easy-to-interview men or hard-to-interview women). Ethnographers adopt two strategies to counter this tendency, which we find useful. First, they make the analysis of these "negative cases" an explicit goal of elicitation. The second, more general strategy is to ask an outsider who is neither a domain expert nor directly involved in the elicitation process to audit or review the elicitor's notes on the domain. Experience has shown that this process can uncover "blind spots" such as verification bias.

When errors are discovered, the knowledge engineer should attempt to determine their cause. Discovering that knowledge is incomplete or only applicable in special contexts is not a cause for concern, but does mean that further elicitation efforts are necessary. If, on the other hand, the expert claims that the knowledge is "wrong," then the knowledge engineer should examine the possibility that the error was produced by faulty or incautious knowledge elicitation. In particular, the knowledge engineer should seek to determine whether there has been a systematic oversimplification of the domain knowledge.

IV. ANALYSIS AND DOCUMENTATION FOR KNOWLEDGE MODELING

The focus of our discussion has been on techniques for eliciting knowledge from an expert. It is assumed that a knowledge engineer takes notes, audio recordings, video recordings, or some combination of the above to preserve a record of the elicitation interviews. Obviously, gathering "raw data" is only the first step in the process. They must then be analyzed and interpreted in a variety of ways to produce a final product. In the four-stage approach we have outlined here, the information gathered in each interview must be analyzed before proceeding to the next. However, a detailed discussion of analysis methods is beyond the scope of this article.

As expressed above, we advocate the documentation of knowledge using what has also been referred to as an intermediate, or mediating representation[21] that can be used in knowledge acquisition without making commitments regarding system implementation tools. Breuker and Wielinga[20] have pointed out the need for such tools, referred to as "knowledge editors," to support knowledge modeling, particularly with the use of bottom-up approaches. We have felt a strong need for such tools to support the approach we have described here. Indeed, we currently have a knowledge editor under development.[42]

V. SUMMARY

We have described a methodology for structuring interviews with domain experts during knowledge elicitation. We have developed our approach by importing and adapting methods from disciplines that have a history of related activities. In doing so, we have been influenced by four major considerations.

First, we have been guided by the nature of expertise, some important aspects of which are (1) a high degree of tightly integrated domain-specific knowledge, (2) problem representations or schema that are based on underlying domain principles and abstractions, (3) general and domain-specific problem-solving strategies present in each problem-solving domain, and (4) a substantial portion of the knowledge is tacit and difficult to articulate.

A second consideration that has influenced our efforts is a concern with having a knowledge engineer avoid reductive bias, one of the pitfalls associated with the acquisition of highly complex concepts. This has led us to place a heavy emphasis on the use of domain-specific language and interviewing methods that attempt to prevent the experts from exercising their translation competence.

A third consideration guiding our development efforts has been the early elicitation of declarative knowledge that is then used to guide the elicitation of procedural knowledge. This is one attempt to address the difficult problem of automaticity in procedural knowledge.

The final consideration influencing our efforts is a conviction that it is advantageous to acquire knowledge independently of concerns for how it will be implemented in a final, working system. This leaves the knowledge engineer free to make decisions about representation and implementation issues without being prematurely constrained in knowledge acquisition efforts.

Given the aforementioned considerations, we have developed a four-phase approach with guidelines for questions suited to each stage. First, the descriptive elicitation stage is intended to reveal the important entities and concepts in the domain as reflected in the terms and specialized language used by the expert. A second stage, structured expansion, is designed to probe the relationships between the domain concepts and the organization of the expert's knowledge using the terminology uncovered in the previous stage. The scripting phase relies on the declarative knowledge found through the two previous stages to discover procedural knowledge, at both macro and micro levels involved in solving domain problems. A final component, validation, is important throughout the process of knowledge elicitation to ensure that the knowledge being elicited is correct and adequate to enable a system in which it is implemented to solve the class problems with it is concerned.

In developing our methodology, we have drawn upon the methods of a number of disciplines, particularly in the social sciences. Other knowledge elicitation theorists[43–45] have recommended similar approaches. We are convinced that field of knowledge elicitation will continue to benefit from this type of interdisciplinary effort.

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